INTRODUCTION TO INFORMATION GEOMETRY

Curves of Distributions and Sampling

Week 4 · Lecture 2

Christopher Lee

A path through probability laws

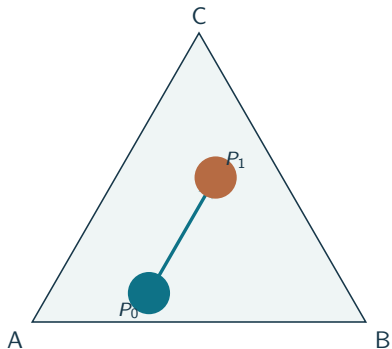
Let P_0 and P_1 be probability vectors. For $0 \leq t \leq 1$,

$$P(t) = (1 - t)P_0 + tP_1.$$

This mixture path stays in the simplex because each coordinate is nonnegative and the coordinates sum to one. Its velocity is constant: $P'(t) = P_1 - P_0$.

A straight path in a triangle

Let $P_0 = (.6, .3, .1)$ and $P_1 = (.2, .3, .5)$.



Every point along the segment is a valid distribution.

The mixture coordinates

For $P_0 = (.6, .3, .1)$, $P_1 = (.2, .3, .5)$, compute

$$P(t) = (1 - t)P_0 + tP_1.$$

Write each component as a function of t and find $P'(t)$.

The mixture coordinates

For $P_0 = (.6, .3, .1)$, $P_1 = (.2, .3, .5)$,

$$P(t) = (.6 - .4t,\ .3,\ .1 + .4t).$$

Its velocity is $P'(t) = (-.4, 0, .4)$.

Differentiate expectation along a path

Fix the outcome values z_i. Along $P(t)$,

$$\mu(t) = \sum_i z_i p_i(t).$$

Differentiating gives

$$\mu'(t) = \sum_i z_i p_i'(t).$$

The value distribution changes because probability moves among the possible outcomes.

The rate of change of a mean

Let $z = (0, 1, 3)$ and $P(t) = (.6 - .4t, .3, .1 + .4t)$.

$$\mu(t) = 0p_A(t) + p_B(t) + 3p_C(t).$$

Find $\mu(t)$ and $\mu'(t)$.

The rate of change of a mean

Let $z = (0, 1, 3)$ and $P(t) = (.6 - .4t, .3, .1 + .4t)$.

$$\mu(t) = .3 + 3(.1 + .4t) = .6 + 1.2t.$$

Thus $\mu'(t) = 1.2$.

What a sample gives us

Suppose n independent observations produce counts n_A, n_B, n_C with total n. The empirical law is

$$\widehat{P} = \left(\frac{n_A}{n}, \frac{n_B}{n}, \frac{n_C}{n} \right).$$

It is itself a point of the simplex. It varies from sample to sample.

An empirical distribution

A sample of 20 observations records

$$n_A = 11, \quad n_B = 6, \quad n_C = 3.$$

Find $\widehat{P}$. Compare it with the proposed model $P = (.6, .3, .1)$.

An empirical distribution

A sample of 20 observations records

$$n_A = 11, \quad n_B = 6, \quad n_C = 3.$$

$$\widehat{P} = (.55, .30, .15).$$

The empirical and model distributions are nearby but unequal.

The expected empirical frequencies

For independent draws from a fixed categorical law P,

$$E[\widehat{p}_i] = p_i.$$

Their variances are

$$\mathrm{Var}(\widehat{p}_i) = \frac{p_i(1-p_i)}{n}.$$

Larger samples typically produce smaller frequency fluctuations.

Sampling coordinates are linked

Because $\widehat{p}_A + \widehat{p}_B + \widehat{p}_C = 1$, an increase in one empirical frequency must be offset elsewhere. For different categories $i \neq j$,

$$\mathrm{Cov}(\widehat{p}_i, \widehat{p}_j) = -\frac{p_i p_j}{n}.$$

The normalization constraint appears in sampling uncertainty as well as in deterministic dynamics.

A comparison of two probability states

Suppose the outcomes are $z = (0, 1, 3)$.

$$P = (.5, .3, .2), \qquad Q = (.3, .3, .4).$$

The change from P to Q transfers 0.2 of the total mass from the outcome 0 to the outcome 3. Its expectation therefore increases by $0.2 \cdot 3 = 0.6$.

Preparing to measure information

We can now describe a distribution, draw samples from it, and follow smooth curves in its state space. Next, we ask two further questions:

- How uncertain is an observation drawn from a given distribution?
- How distinguishable are two nearby distributions?

The first leads to entropy; the second leads toward information geometry.